

Task 4: Compare Results and Reflect

What differences did you observe?

I noticed that the baseline model, which was trained on pre-collected data from other users, had a more general understanding of digit gestures. After fine-tuning with my own collected data, the model became more adapted to my specific drawing style, including how I size each digit, my drawing speed, and the angle at which I hold the Arduino.

Which model performed better and why?

The fine-tuned model performed better on my personal gestures because it was specifically trained on my own drawing style. The baseline model struggled with some of my digits since it was trained on other people's data and didn't account for my particular way of drawing certain digits. Fine-tuning essentially personalized the model to my gesture style.

Did fine-tuning help for your specific gesture style?

Yes, fine-tuning did help for my specific gesture style. After fine-tuning, the model was better at recognizing the digits I drew in the air. However, since my collected dataset was missing some digits and had fewer samples for others, the improvement was not uniform across all digits. Digits where I had more clean samples showed the most improvement, while digits with fewer samples still showed some misclassifications.